



e-Courier

Hub-and-Spoke Courier Management System — Complete Workflow Guide

Covers Admin, Hub Manager, Merchant & Rider dashboards
End-to-end parcel lifecycle · Bags · Trips · COD · Returns · API

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1 · System Overview & Core Concepts

e-Courier is a **hub-and-spoke** courier platform (SteadFast-style) for parcel delivery, currently operating in **Dhaka**. Parcels never move merchant-to-customer directly — they flow through **hubs**, and every movement is recorded.

The four roles / dashboards

Role	Who they are	What they do
Admin (HQ)	Head office	Oversees everything: merchants, all parcels, hubs, zones/pricing, COD, payments, support, reports, branding.
Hub Manager	Runs one hub	Pickups, bagging, trips, areas, rider management, COD for their hub.
Merchant	Online seller	Books parcels, tracks them, manages COD, integrates the API.
Rider (Delivery man)	Field staff	Runs trips: picks up from merchants, delivers to customers, collects COD.

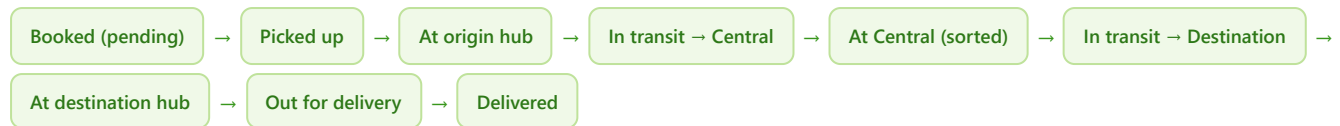
The three hub types

Hub	Role
Origin hub	The local hub for the merchant's area — riders bring picked-up parcels here.
Central hub	The sorting facility — every inter-area parcel routes through it and is re-sorted by destination.
Destination hub	The local hub for the customer's area — last-mile riders deliver from here.

A hub is both an origin (for its merchants) and a destination (for its customers). Routing is always **origin** → **central** → **destination**. If origin = destination, it skips central.

Key rule — everything is a scan. Each physical/logical movement writes a **scan event** (booked, pickup, hub-inbound, hub-outbound, out-for-delivery, delivered, failed, return) tagged with the hub. A parcel's status and location are always derived from this scan trail, giving a complete audit history you see on every tracking timeline.

2 · The Parcel Lifecycle (end-to-end)



Statuses used

Status	Meaning	Set when
pending	Booked, awaiting pickup	Merchant books it
picked_up	Collected from the merchant	Rider picks it up on a trip
at_hub	Resting at a hub	Submitted to hub / received via a bag
in_transit	Moving between hubs (in a bag)	A bag is dispatched
out_for_delivery	On a rider's delivery trip	Destination hub's rider starts a trip
delivered / partially_delivered	Handed to customer, COD collected	Rider delivers (OTP verified)
return_in_transit → returned	Reversing back to the merchant (RTO)	Failed delivery / return started
cancelled	Cancelled before pickup	Merchant cancels a pending parcel

COD (Cash on Delivery)

On delivery the rider collects the COD amount. At the end of a trip the rider hands cash to the hub, which is **reconciled** against the COD that should have been collected (short amounts are recorded as owed). The hub remits to HQ; HQ disburses to the merchant minus charges.

3 · Roles & Data Visibility (who sees what)

Visibility follows the chain of custody so each screen shows only relevant parcels:

Viewer	Sees
Admin	Everything, across all hubs and merchants.
Merchant	Only their own parcels. The delivery rider's identity is shown only at final delivery (never the pickup rider).
Origin hub	Its outbound parcels from booking onward (it owns them), even after handing off — for tracking.
Central hub	The whole active in-transit pipeline routed through it.
Destination hub	A parcel only once it is dispatched toward it (in the incoming queue), then while it holds it.
Rider	Only parcels assigned to them.

A hub can **see** a handed-off parcel but can only **act** on parcels it currently holds or that are inbound to it.

4 · Getting Started & Login Accounts

Open the app, go to **Login**. Demo quick-fill buttons are on the login page. Default rider password is `12345678` .

Role	Email	Password
Admin (HQ)	admin@cms.com	admin123
Merchant	merchant@cms.com	merchant123
Central Hub	central@cms.com	central123
Mirpur Hub	mirpur@cms.com	mirpur123
Gulshan Hub	gulshan@cms.com	gulshan123
Rider	(created by a hub manager)	12345678

The app auto-routes each account to its own dashboard after login. Public pages (landing, **Track** a parcel) need no login.

5 · Admin Dashboard — see & manage

HQ oversight of the whole operation. Left-sidebar navigation:

Page	See	Manage
Dashboard	Totals, recent parcels, volume chart	—
Merchants	All merchants + COD balances	Approve/suspend, edit, assign home hub, open detail
Parcels	Every parcel; filter by status, merchant, zone, hub , date; Location column	Open detail, bulk status update, export CSV
Pickup Requests	Merchant pickup requests	Process
Delivery Men	All riders	Open detail, edit, reset password, activate/deactivate
Hubs	All hubs + coverage	Create, edit, set coverage, activate/deactivate
Zones	Pricing zones	Add / edit / delete zone & charges
COD	COD collected/pending per merchant	Reconcile / disburse
Payments	Merchant withdrawals	Approve / mark paid
Support	All support tickets	Reply, set status/priority
Reports	Operational analytics	—
Settings	Account	Change password; Company & Branding (name, logo, contact — shows across the whole site)

6 · Hub Manager Dashboard — see & manage

Runs the day-to-day of one hub. Everything is scoped to that hub.

Page	See	Manage
Dashboard	Hub totals, recent parcels at the hub	—
Pickup Requests	Incoming pickup requests	Assign a pickup rider
Parcels	Tabs: Incoming (accept/reject), At Hub , Outbound , All, Delivered, Returned. Shows "last action by"	Accept / reject inbound, assign delivery rider, dispatch, Return to sender
Bags	Incoming bags, ready-to-bag parcels grouped by next hub, outbound bags	Build bag, dispatch bag, receive & break bag
Trips	Riders on the road, closed trips, outstanding rider dues (COD shortfalls)	Oversight / reconciliation view
Areas	Hub coverage areas	Add / edit / delete areas (cascades to riders)
Delivery Men	Riders at this hub	Add (with areas), open detail, edit (details + areas), reset password, activate/deactivate
COD	Cash collected/handed over at the hub	Confirm rider handovers, remit to HQ
Reports	Hub analytics	—

7 · Merchant Dashboard — see & manage

Page	See	Manage
Dashboard	Balance, parcel counts, ongoing parcels, 7-day volume	Row-click a parcel to open it
Book Parcel	Booking form with live charge preview	Create a parcel (recipient, area, COD, weight)
My Parcels	All their parcels; filter/search/export	Row-click to open (View); track
Import	Bulk CSV importer	Import many parcels
Pricing	Zone charges / calculator	Estimate delivery cost
COD	COD collected/pending	—
Payments	Withdrawal history	Request withdrawal
Reports	Their analytics	—
Developers	API credentials, docs, webhook config, delivery log	Create/revoke Api-Key + Secret-Key; set Callback URL + Auth Token
Support	Their tickets	Open a ticket, chat
Profile	Shop info	Edit profile

On a parcel's detail, the merchant sees status milestones (incl. "Picked Up") and — only once it reaches final delivery — the assigned delivery rider's name & phone.

8 · Rider App — see & manage

Mobile-first. Bottom tabs: Home · Trip · Parcels · Cash · Profile. The **Trip** is the single place all rider actions happen.

Tab	See	Manage
Home	Counts (To Pick Up, To Deliver, Delivered, Refund), Up Next	—
Trip	Start screen (parcels ready) or the active runsheet: Deliveries + Pickups	Start trip; Deliver (OTP, full/partial); Fail (reason); Pick up; Close trip (hand in cash → reconciliation)
Parcels	Read-only list of their parcels (Active / Delivered / Closed)	Open a parcel for read-only info + tracking
Cash	Cash in hand / handed over	—
Profile	Their details	Change password

Real-time: a rider gets a live push (bell + list) the instant a parcel is assigned — pickups show the merchant's pickup address; deliveries show the recipient.

9 · End-to-End Workflows (step by step)

A · Full journey: book → deliver (cross-role)

- 1 **Merchant** books a parcel (Mirpur→Gulshan). Status `pending` , at Mirpur.
- 2 **Mirpur hub** assigns a pickup rider (Parcels or Pickup Requests).
- 3 **Mirpur rider** opens a Trip → adds the pickup → status `picked_up` ; on **Close trip** it's submitted to the hub → `at_hub` at Mirpur.
- 4 **Mirpur hub** → **Bags**: builds a bag to Central, dispatches it → `in_transit` .
- 5 **Central hub** → **Bags**: receives & breaks the bag → parcels `at_hub` at Central; re-bags by destination → dispatches to Gulshan.
- 6 **Gulshan hub** → **Bags**: receives & breaks → parcels `at_hub` at Gulshan (ready for delivery).
- 7 **Gulshan hub** → **Parcels**: assigns a delivery rider.
- 8 **Gulshan rider** starts a Trip (parcels go `out_for_delivery`), delivers with OTP → `delivered` , COD collected.
- 9 **Close trip**: rider hands in cash → reconciled; hub sees it under **Trips**.

B · Bags / line-haul (hub manager)

- **Build**: on **Bags**, parcels ready to move are grouped by next hub — "Create bag".
- **Dispatch**: sends the bag (all parcels → in transit together).
- **Receive & break**: the receiving hub scans the bag in (all parcels → at hub). Central then re-bags by destination.

C · Zone-rider Trip + COD reconciliation (rider + hub)

- One rider, one round-trip: **deliver** (collect COD) + **pick up** new parcels in their zone.
- **Close trip**: enter cash handed in. System shows COD due vs collected; shortfalls are recorded as **rider dues** the hub tracks under **Trips**.

D · Return to sender (RTO)

- After failed attempts, the hub uses **Return to sender** on the parcel.
- It reverses through the chain (destination → central → origin) via bags and completes as `returned` at the origin hub.

E · Areas & rider assignment (hub manager)

- **Areas** page: add / rename / delete the hub's delivery areas (renames & deletes cascade to riders).
- Assign a rider's areas directly in the rider's **Add** or **Edit** form (chips of the hub's areas).

F · Merchant API (Steadfast-style)

- Merchant → **Developers**: create credentials (Api-Key + Secret-Key, secret shown once).
- Their site calls `POST /api/merchant/v1/create_order` (single) or `/create_order/bulk-order` with those headers.
- Set a **Callback URL + Auth Token** to receive live `delivery_status` / `tracking_update` webhooks.

G · Branding (admin)

- Admin → **Settings** → **Company & Branding**: set company name, logo, contact.
- These appear across the sidebar, login, public landing/footer, tracking page and shipping labels.

10 · Data Model Summary

Entity	Purpose
Parcel	The shipment: recipient, COD, charges, origin/current/destination hubs, status, assigned rider, returning flag.
Scan event (history)	One row per movement (type + hub + who + when) — the tracking timeline.
Hub (Branch)	Origin / central / destination; has coverage areas.
Bag	A manifest of parcels for one hub→hub transfer (open → dispatched → received).
Trip + Trip items	A rider's round-trip: delivery + pickup items, expected vs collected COD.
Delivery man (Rider)	Belongs to a hub; covers specific areas.
Merchant	Seller; homed at a hub; API credentials + webhook.
Zone	Pricing (delivery/COD/return charges) by district.
COD handover / remittance	Cash chain: rider → hub → HQ → merchant.
Site settings	Company name, logo, contact (branding).

11 · Deployment & Operations Notes

- **Stack:** Django REST + Channels (WebSockets) backend; Next.js frontend.
- **Dev ports:** backend `:8100` , frontend `:3100` .
- **Real-time:** served under ASGI (Daphne). In dev the channel layer is in-memory (single process). For multi-worker production, switch `CHANNEL_LAYERS` to Redis (`channels_redis`) so live notifications work across workers. REST notifications work either way.
- **Merchant API rate limits & idempotency** use the Django cache — use a shared cache (Redis) in production.
- **Coverage:** launch is Dhaka-only; add hubs + areas to expand. An area belongs to exactly one hub.
- **Media:** logos, rider photos/documents, and delivery proof images are uploaded files — configure media storage in production.

Verification status: full backend lifecycle test 33/33 passing; browser UI sweep 40 pages with 0 runtime errors; notifications (REST + real-time) verified.